ESKAN
DDS MESSAGE PROTOCOL

Message Protocol v12

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Eskan

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1 Overview

This document describes the serial UART communication interface used on the DDS EORF0013AA10 module.

The DDS has a dumb frequency multiplier on its output. This provides frequency multiplication options of x1, x2, x4, x8 appropriate filtering and gain for +10dBm output.

This multiplication factor is stored in the onboard EE.

The DDS is one of potentially multiple slave devices on the UART bus, each having a different unit ID, also configured in the onboard EE.

Communication is done as a serial bit stream using 38.4kbps, 8 data, 1 stop, no parity.

Communication is initiated by the controlling master and proceeds message by message. Valid commands addressed to the unit normally return ACK, NAK or DATA as specified for each command. The global Set Unit ID command and successful reset or bootloader-entry commands do not return a response. Frames with invalid CRCs and messages for other unit IDs are ignored. Each message is packed in a frame to identify its start and end.

2 Substitution Protocol

2.1 Substitution

Any byte with value 0x01 in the message, including either CRC byte, is substituted with 0x06 0x21.

Any byte with value 0x02 in the message, including either CRC byte, is substituted with 0x06 0x22.

Any byte with value 0x06 in the message, including either CRC byte, is substituted with 0x06 0x26.

The frame starts with 0x01 and ends with 0x02. These frame markers are not substituted. Calculate and append the CRC before substituting bytes throughout the message (unit ID, command ID, payload and CRC). Each original byte is substituted once; the inserted 0x06 escape byte is not substituted again. On reception, remove the frame markers and reverse the substitutions before validating the CRC.

2.2 Structure

The block inside the frame is the message. It has following structure:

Byte 0: Module ID is the assigned address of the slave. Address 0xFF is reserved for global messages. The current application firmware acts on a global message only for Set Unit ID (0xA5) with a new ID other than 0xFF. All connected slaves accept that new ID and send no response, avoiding bus collisions. Other broadcast commands are ignored.

Byte 1: Message ID defines the kind of message as well as the structure of following bytes in the payload.

Byte 2 ... Payload as defined by the Command ID.

The message then ends by appending its two-byte CRC-16.

2.3 CRC

Calculate the CRC-16 over the original message bytes in order: unit ID, command ID and any payload. Exclude the start and end frame markers and the CRC itself. Calculate the CRC before byte substitution; on reception, reverse the substitutions before validating it.

CRC settings: width 16 bits; polynomial 0x1021, f(x) = x^16 + x^12 + x^5 + 1; initial value 0x0000; reverse the bit order within each input byte; do not reverse the output; final XOR 0x0000. For a software implementation, start with zero, reverse each input byte, XOR it into the high byte of the register, then perform eight left shifts, XORing 0x1021 whenever the outgoing most significant bit is 1. Retain only 16 bits after each shift.

Append the resulting CRC least significant byte first, then most significant byte. Apply byte substitution to the entire message including both CRC bytes. For example, the bytes 0x20 0xB2 0x00 0x01 0x00 produce CRC 0xC584, appended as 0x84 0xC5. The receiver recalculates the CRC over the decoded message excluding its final two bytes and compares it with those bytes in the same order. Messages with a CRC mismatch are ignored.

2.4 Message IDs

Message IDs above and including 0xA0 are intended to provide a common interface across all slave products. Message IDs below 0xA0 are for test, development, and specific user commands.

For example, the first data byte in the status message 0xA0 contains a bitfield with a summary of overall status/warnings and alarm state, the following bytes are module specific. This should be emulated.

2.5 Example

In this example the master sets the version number of slave (Unit ID 0x20) to 1.

Tx: Command sent by master, Rx: Response from Slave.

Tx payload: 0x20 0xB2 0x00 0x01 0x00

Tx message: 0x01 0x20 0xB2 0x00 0x06 0x21 0x00 0x84 0xC5 0x02

Rx message: 0x01 0x20 0xB2 0x41 0x9A 0x16 0x02

Rx payload: 0x20 0xB2 0x41

The structure of the payloads is described below.

2.5.1 Master Command - Payload Explanation

0x20 0xB2 0x00 0x01 0x00

0x20: Slave unit ID (destination address).

0xB2: Command ID, Set PCB Version.

0x00: Build information target index. 0 selects the main DDS board; 1 selects the multiplier board. Other indices are not supported.

0x01: Least significant byte of the 16-bit PCB version.

0x00: Most significant byte of the PCB version. The two bytes 0x01 0x00 represent version 1 (0x0001), transmitted least significant byte first.

2.5.2 Master Command - Message Explanation

The payload is 0x20 0xB2 0x00 0x01 0x00.

The first step is to calculate and append the CRC-16, least significant byte first.

0x20 0xB2 0x00 0x01 0x00 0x84 0xC5

The second step is to substitute 0x01, 0x02 and 0x06 in the payload and CRC. Here, 0x01 becomes 0x06 0x21.

0x20 0xB2 0x00 0x06 0x21 0x00 0x84 0xC5

The third step is to add the Start of Message and end of message identifiers.

0x01 0x20 0xB2 0x00 0x06 0x21 0x00 0x84 0xC5 0x02

Sends message to Slave.

2.5.3 Slave receives message.

0x01 0x20 0xB2 0x00 0x06 0x21 0x00 0x84 0xC5 0x02

The first step for the slave is to remove the start and stop identifiers.

0x20 0xB2 0x00 0x06 0x21 0x00 0x84 0xC5

The second step is to reverse the byte substitutions in the payload and CRC.

Here, 0x06 0x21 becomes 0x01. The decoded message, including its CRC, is:

0x20 0xB2 0x00 0x01 0x00 0x84 0xC5

The third step is to validate the CRC encoded in the last two decoded bytes, least significant byte first.

The CRC calculated over 0x20 0xB2 0x00 0x01 0x00 is 0xC584, matching the received bytes 0x84 0xC5. If the CRC does not match, the slave ignores the message. The payload is:

0x20 0xB2 0x00 0x01 0x00

Only if this slave device has a unit ID of 0x20 will the slave device action the message, by setting the version to 1. If this is successful reply with an ACK, else a NAK.

2.5.4 Slave Acknowledges message, ACK.

Payload is: 0x20 0xB2 0x41.

0x20: Slave unit Id

0xB2: Command ID replying to.

0x41: Response Type, 0x41: ACK message, instruction carried out.

The first step is to calculate the CRC-16 over 0x20 0xB2 0x41 and append it least significant byte first. The CRC is 0x169A, appended as 0x9A 0x16.

0x20 0xB2 0x41 0x9A 0x16

The second step is to apply byte substitution to the payload and CRC. No substitution is required in this example.

The third step is to add the start of message and end of message identifiers.

0x01 0x20 0xB2 0x41 0x9A 0x16 0x02

Send response to Master.

2.5.5 Slave Error message, NAK.

If the EE write fails, the slave will respond with a NAK and error code 101.

Payload: 0x20 0xB2 0x4E 0x65 0x00.

0x20: Slave unit Id

0xB2: Command ID replying to.

0x4E: Response Type, 0x4E: NAK, Not Acknowledge, instruction not carried out.

0x65: Error number LSB

0x00: Error number MSB

The first step is to calculate the CRC-16 over 0x20 0xB2 0x4E 0x65 0x00 and append it least significant byte first. The CRC is 0xDFAC, appended as 0xAC 0xDF.

0x20 0xB2 0x4E 0x65 0x00 0xAC 0xDF

The second step is to apply byte substitution to the payload and CRC. No substitution is required in this example.

0x20 0xB2 0x4E 0x65 0x00 0xAC 0xDF

The third step is to add the Start of Message and end of message identifiers.

0x01 0x20 0xB2 0x4E 0x65 0x00 0xAC 0xDF 0x02

Send response to Master.

3 Response Types

The third byte (byte 2 if counting from byte 0) in the payload contains the response type.

3.1 Response Type 0x44: Data to follow.

This will be specific to the command, see command/response.

3.2 Response Type 0x41: ACK, Acknowledge

ACK Response:

0	0x20	Unit ID
1	0xHH	Command ID replying to
2	0x41	Response type, 0x41: ACK, Acknowledge, Successful

3.3 Response 0x4E: NAK, Not Acknowledge

NAK Response:

0	0x20	Unit ID
1	0xHH	Command ID responding to
2	0x4E	Response type, 0x4E: NAK, Not successful
3	0xnn	Error number LSB
4	0xnn	Error number MSB

Error Number Table:

100	Value OOR (Out of Range)
101	EE_W, EE Write via I2C fail
102	EE_R, EE Read via I2C fail
103	Sweep Mode OOR (Out of Range)
104	Barcode unknown
105	CW Freq OOR
106	Unknown Command
107	Index OOR
108	Number of sweeps OOR
109	SWP 1 Start Frequency OOR
110	SWP 1 Stop Frequency OOR

111	SWP 2 Start Frequency OOR
112	SWP 2 Stop Frequency OOR
113	SWP 3 Start Frequency OOR
114	SWP 3 Stop Frequency OOR
115	SWP 4 Start Frequency OOR
116	SWP 4 Stop Frequency OOR
117	SWP 5 Start Frequency OOR
118	SWP 5 Stop Frequency OOR
119	SWP 6 Start Frequency OOR
120	SWP 6 Stop Frequency OOR
121	Frequency Step OOR
122	Frequency Step OOS (Out of Step)
123	Dwell time OOR
124	Modulation time OOR (Out of Range)
125	Modulation deviation OOS (Out of Step)
126	Modulation pattern OOR (Out of Range)
127	Modulation frequency selection OOR (Out of Range)
128	Sweep too fast for the selected modulation mode
129	Dither unsupported for the requested sweep configuration

4 Commands and Responses

4.1 Command 0x40: Read Voltages.

Reports the 3V3 supply estimate, 1V8 power-good indication and raw internal-reference calibration data for unit ID 0x20.

Command:

0	0x20	Unit ID
1	0x40	Message ID

Response:

0	0x20	Unit ID
1	0x40	Message ID
2	0x44	Data to follow
3	0x05	3V3 Supply in mV 0x0D05 = 3333mv
4	0x0D	
5	0x08	1V8 power-good indication (16-bit, least significant byte first) 0x0708 = 1800 when PG_1V8 is high; 0x0000 when low. This is not a measured voltage.
6	0x07	
7	0x1C	Raw VREFINT calibration value (16-bit, least significant byte first) Example: 0x0C1C = 3100 counts. The actual calibration value is device-dependent; this is not a 3V3P voltage in mV.
8	0x0C	

The 1V8 field is derived from the PG_1V8 pin: 1800 when high, otherwise 0. It is not an ADC measurement. The field previously labelled 3V3P currently contains the factory VREFINT calibration value read from VREFINT_CAL_ADDR, not a measured 3V3P voltage. All three fields are 16-bit values transmitted least significant byte first. The example values are illustrative.

4.2 Command 0x50: Set EE to Default Settings.

Sets the config held in EE to a known default state, suitable for test.

Default settings	
Bar code string	EJWM0000AA
PCB version	0
SN string	TBD0000000
Multiplier Factor	1
Min Freq (KHz)	100,000
Max Freq (KHz)	900,000
Temperature Alarm Threshold (deg. C)	60
Output Level	78
Sweep Mode	1: Sweep
Number of sweeps	1
Sweep 1, Start Freq (KHz)	600,000
Sweep 1, Stop Freq (KHz)	625,000
Sweep Step Frequency (Hz)	1000
Ramp Rate	20
Modulation time	1 s
Modulation pattern	Off
Modulation frequency selection	100 kHz

Command:

0	0x20	Unit ID
1	0x50	Message ID

Response:

0	0x20	Unit ID
1	0x50	Message ID
2	0x44	Data to follow
3	0x31	0x31=DONE, 0x30=FAIL

If an EE write fails, the command returns a DATA response (0x44) with result 0x30 (FAIL). A successful command returns result 0x31 (DONE).

The unit ID is preserved, and restarting (power cycle or send STM32 reset command) remains necessary to reload the complete configuration. Writes are sequential, so a failure can leave some defaults already written.

4.3 Command 0x51: EE Erase.

Writes 0xFF to EEPROM addresses 0x0000 to 0x00FF inclusive (256 bytes), in sixteen 16-byte blocks. This command does not erase the remainder of the EEPROM.

The current unit ID remains active until a power cycle or STM32 reset. After a successful erase and restart, the stored unit ID is 0xFF, which the firmware maps to unit ID 0xFD if the EEPROM read succeeds. If the unit ID cannot be read, the firmware uses 0xFE.

Command:

0	0x20	Unit ID
1	0x51	Message ID

Response:

0	0x20	Unit ID
1	0x51	Message ID
2	0x44	Data to follow
3	0x31	0x31=DONE, 0x30=FAIL

The command returns a DATA response (0x44) with result 0x31 (DONE) on success or 0x30 (FAIL) if an EE write fails. Erasing stops at the first failed write, so earlier blocks may already have been erased.

4.4 Command 0x52: EE Read.

Reads the EE data at address (0->0xFF) of unit ID 0x20.

Command:

0	0x20	Unit ID
1	0x52	Message ID
2	0x20	EE Address = 0x20

Response:

0	0x20	Unit ID
1	0x52	Message ID
2	0x44	Data to follow
3	0x45	EE Data = 0x45

NAK response with error 102 (EE read failure) if the EEPROM read fails.

4.5 Command 0x53: EE Write.

Writes EE data to address (0->0xFF) of unit ID 0x20.

Command:

0	0x20	Unit ID
1	0x53	Message ID
2	0x20	EE Address = 0x20
3	0x45	EE Data = 0x45

Response:

ACK on successful write; NAK with error 101 (EE write failure) if the EEPROM write fails.

4.6 Command 0x54: EE Test.

Tests EEPROM addresses 0x0000 to 0x00FF inclusive (256 bytes) on unit ID 0x20. The test first writes the 8-bit bitwise complement of each address to that location (for example, 0xFF at address 0x00 and 0x00 at address 0xFF), then reads the range back and checks each byte. Original contents are overwritten and are not restored.

A complete test takes several seconds; duration depends on EEPROM and I2C operation. It stops at the first write failure, read failure or data mismatch and returns a DATA response (0x44) with result 0x30 (FAIL). Successful completion returns result 0x31 (PASS). If a write fails, only part of the range may have been overwritten.

The tested range includes stored configuration and the unit ID. A complete write pass sets the stored unit ID at address 0x10 to 0xEF. The current unit ID remains active until a power cycle or STM32 reset, when the stored ID is reloaded if the EEPROM read succeeds.

Command:

0	0x20	Unit ID
1	0x54	Message ID

Response:

0	0x20	Unit ID
1	0x54	Message ID
2	0x44	Data to follow
3	0x31	0x30 = FAIL, 0x31 = PASS

4.7 Command 0x60: Get STM32 Pin State.

Request the state of some uC hardware control signals on unit ID 0x20.

Command:

0	0x20	Unit ID
1	0x60	Message ID

Response:

0	0x20	Unit ID
1	0x60	Message ID
2	0x44	Data to follow
3	0x30	LED1 (GRN), 0x30: OFF / 0x31: ON
4	0x30	LED2 (BLUE), 0x30: OFF / 0x31: ON
5	0x30	EN_3V3P, 0x30: OFF / 0x31: ON

Note: Some of these signals are changing dynamically.

4.8 Command 0x61: Set Green LED ON/OFF.

This command allows low level setting of the green LED1 to ON or OFF.

Command:

0	0x20	Unit ID
1	0x61	Message ID
2	0x31	0x31=ON, 0x30=OFF

Response:

ACK when the control byte is 0x31 (ON) or 0x30 (OFF); NAK with error 100 (out of range) for any other control byte.

4.9 Command 0x62: Set Blue LED ON/OFF.

This command allows low level setting of the blue LED2 to ON or OFF.

Command:

0	0x20	Unit ID
1	0x62	Message ID
2	0x31	0x31=ON, 0x30=OFF

Response:

ACK when the control byte is 0x31 (ON) or 0x30 (OFF); NAK with error 100 (out of range) for any other control byte.

4.10 Command 0x63: 3V3P ENABLE.

This command allows low level setting of the 3V3P supply dedicated to I2C devices to ON or OFF.

Command:

0	0x20	Unit ID
1	0x63	Message ID
2	0x31	0x31=ON, 0x30=OFF

Response:

ACK when the control byte is 0x31 (ON) or 0x30 (OFF); NAK with error 100 (out of range) for any other control byte.

4.11 Command 0x64: Set Low Level Test Mode.

This command enables or disables the low level test-mode flag. Control byte 0x31 enables the flag and sets the test-mode warning bit; 0x30 disables the flag and clears that warning bit. The current firmware does not use this flag to isolate the LEDs from their normal functions.

Command:

0	0x20	Unit ID
1	0x64	Message ID
2	0x31	0x31 = Enable / 0x30 = Disable

Response:

ACK when the control byte is 0x31 (enable) or 0x30 (disable); NAK with error 100 (out of range) for any other control byte.

4.12 Command 0x70: Get DDS Register Value

See AD9915 datasheet for register details.

Reads a 16-bit value from DDS register address 0x00 to 0x6F on unit ID 0x20. The response sends the least significant byte first, followed by the most significant byte. The table shows an example value of 0x0800; the actual value depends on the current register contents.

Command:

0	0x20	Unit ID
1	0x70	Message ID
2	0x6D	Register Address 0x6D

Response:

0	0x20	Unit ID
1	0x70	Message ID
2	0x44	Data to follow
3	0x00	Data LSB (example: 0x00 for value 0x0800)
4	0x08	Data MSB (example: 0x08 for value 0x0800)

NAK with error 100 (out of range) if the register address is above 0x6F.

4.13 Command 0x71: Set DDS register Value.

See AD9915 datasheet for register details.

This command is intended to set a DDS register value on unit ID 0x20, but register writing is currently unimplemented. The firmware checks only whether the address is in the range 0x00 to 0x6F. It ignores the supplied data bytes and does not change the DDS register.

Command:

0	0x20	Unit ID
1	0x71	Message ID
2	0x6D	Register Address 0x6D
3	0x00	Data LSB (example: 0x00 for 0x0800); currently ignored
4	0x08	Data MSB (example: 0x08 for 0x0800); currently ignored

Response:

ACK for register addresses 0x00 to 0x6F; this confirms only that the address is accepted, not that a write occurred. NAK with error 100 (out of range) if the register address is above 0x6F.

4.14 Command 0x72: Test DDS registers.

DDS register testing is currently unimplemented. This command performs no DDS register test and always returns a DATA response (0x44) with result 0x30 (FAIL). The response format defines 0x31 as PASS, but the current implementation never returns it.

Command:

0	0x20	Unit ID
1	0x72	Message ID

Response:

0	0x20	Unit ID
1	0x72	Message ID
2	0x44	Data to follow
3	0x30	0x30: FAIL (always returned); 0x31: PASS (not returned by current firmware)

4.15 Command 0x80: Set Sweep Config. (User)

4.15.1 Command to configure a CW frequency:

Command:

0	0x20	Unit ID
1	0x80	Message ID
2	0x00	Mode = 0: CW, 1: Sweep
3	0x50	CW output frequency in kHz, 32-bit unsigned, least significant byte first. Example: 0x00256250 = 2,450,000 kHz. The frequency must lie within the configured limits; this example is outside the 100,000 to 900,000 kHz default limits.
4	0x62	
5	0x25	
6	0x00	
7	0x00	Modulation mode: 0 = None, 1 = Dither, 2 = Injected. Required even when modulation is disabled.
8	0x20	Packed modulation settings. Example 0x20 selects 1 s, pattern OFF and 100 kHz. Required even when modulation is disabled.

Byte 8 bits 1-0 select modulation time: 0 = 1 s, 1 = 100 ms, 2 = 100 us, 3 = random.

Byte 8 bits 3-2 select the pattern: 0 = OFF, 1 = PRBS3, 2 = PRBS5, 3 = PRBS7. Bits 5-4 select modulation frequency: 0 = OFF, 1 = 75 kHz, 2 = 100 kHz, 3 = 150 kHz. Set unused bits 7-6 to zero.

The table shows the compact CW command format. Append the two CRC bytes after byte 8, then apply byte substitution and frame markers as described in section 2. The decoded message is 11 bytes including the CRC.

Response:

ACK after the configuration is saved to EEPROM and applied to the DDS. NAK with error 105 if the CW frequency is outside the configured minimum and maximum limits; error 100 if required modulation fields are missing or the modulation mode is invalid; error 101 if an EEPROM write fails.

4.15.2 Command to configure the sweeps:

Up to 6 sweeps can be configured in a band. This example assumes a configured x4 frequency multiplier and output frequency limits of 2,000,000 to 4,000,000 kHz (2 to 4 GHz). It uses three sweeps between 2.4 and 2.5 GHz. The default test setup uses x1 and different limits.

Set byte 3 (number of sweeps) to 0 to disable all sweeps, or to 1 through 6 to select the number of active sweeps. Ramp rate and modulation fields must still contain valid values when the count is zero.

Include all six start/stop frequency pairs, even when fewer than six sweeps are active. Pairs beyond the selected count are not validated as active sweeps; use zero for these unused pairs as shown.

Start/stop frequencies are 32-bit unsigned output frequencies in kHz, transmitted least significant byte first. The firmware divides them by the configured multiplier for the DDS. Active sweeps must be ordered from lowest to highest frequency and lie within the configured output limits.

0	0x20	Unit ID
1	0x80	Message ID
2	0x01	Mode 0: CW, 1: Sweep
3	0x03	Number of Sweeps = 0 to 6
4	0x00	Sweep 1 Start Frequency 0x249F00 2400,000 KHz
5	0x9F	
6	0x24	
7	0x00	
8	0x20	Sweep1 Stop Frequency 0x24ED20 2420,000 KHz
9	0xED	
10	0x24	
11	0x00	
12	0x40	Sweep 2 Start Frequency 0x253B40 2440,000 KHz
13	0x3B	
14	0x25	
15	0x00	
16	0x60	Sweep 2 Stop Frequency 0x258960 2460,000 KHz
17	0x89	
18	0x25	
19	0x00	
20	0x80	Sweep 3 Start Frequency 0x25D780 2480,000 KHz
21	0xD7	
22	0x25	
23	0x00	
24	0xA0	Sweep 3 Stop Frequency 0x2625A0 2500,000 KHz
25	0x25	
26	0x26	
27	0x00	
28	0x00	Sweep 4 Start Frequency

29	0x00	
30	0x00	
31	0x00	
32	0x00	Sweep 4 Stop Frequency
33	0x00	
34	0x00	
35	0x00	
36	0x00	Sweep 5 Start Frequency
37	0x00	
38	0x00	
39	0x00	
40	0x00	Sweep 5 Stop Frequency
41	0x00	
42	0x00	
43	0x00	
44	0x00	Sweep 6 Start Frequency
45	0x00	
46	0x00	
47	0x00	
48	0x00	Sweep 6 Stop Frequency
49	0x00	
50	0x00	
51	0x00	
52	0x88	Sweep frequency step in Hz, 32-bit unsigned, least significant byte first. Example: 0x00001388 = 5,000 Hz. Valid range: 1,000 to 4,000,000 Hz when at least one sweep is active.
53	0x13	
54	0x00	
55	0x00	
56	0x14	Ramp Rate, 32-bit unsigned, least significant byte first. Example: 0x00000014 = 20. Valid range: 1 to 65,535 DDS ramp-rate counts; this is not a time in nanoseconds.
57	0x00	
58	0x00	
59	0x00	
60	0x00	Modulation mode: 0 = None, 1 = Dither, 2 = Injected.
61	0x20	Packed modulation settings, as defined for byte 8 in section 4.15.1: bits 1-0 = time, bits 3-2 = pattern, bits 5-4 = modulation frequency. Set bits 7-6 to zero. Example 0x20 = 1 s, pattern OFF, 100 kHz.

For sweep 1, start must be at least the configured minimum and no greater than the maximum; stop must be at least start + floor(step Hz / 1000) kHz and no greater than the maximum. For sweeps 2 to 6, each active start must be at least the preceding stop and no greater than the maximum; each active stop must be strictly greater than start + floor(step Hz / 1000) kHz and no greater than the maximum.

Dither and injected modulation impose additional sweep-speed limits. Dither also requires a supported sweep configuration; the firmware rejects settings that fail these checks. Both modulation bytes are required even when modulation is disabled.

Append the two CRC bytes after byte 61, then apply byte substitution and frame markers as described in section 2. Include all fields regardless of the number of active sweeps. The decoded message is 64 bytes including the CRC.

Response:

Success returns ACK after the configuration is saved to EEPROM and applied to the DDS. Failure returns NAK with an error listed below. See section 3.3 for the complete error-number table.

NAK errors

Error code	Meaning	Trigger
100	Invalid modulation mode	Byte 60 is greater than 2. Valid values are 0 (None), 1 (Dither) and 2 (Injected).
101	EEPROM write failure	An EEPROM write fails while saving the configuration.
108	Invalid sweep count	Byte 3 is greater than 6.
109-120	Invalid sweep start or stop	An active sweep fails the frequency-limit, ordering or minimum-span checks described above. Codes are start/stop pairs for sweeps 1 to 6 respectively: 109/110 through 119/120.
121	Frequency step out of range	At least one sweep is active and the step is below 1,000 Hz or above 4,000,000 Hz.
123	Ramp Rate out of range	The ramp-rate count is below 1 or above 65,535.
128	Sweep too fast	The step and ramp-rate combination fails the timing check for the selected Dither or Injected modulation mode.
129	Unsupported dither configuration	Dither mode is selected and the requested sweep configuration cannot use a supported dither plan.

4.16 Command 0x81: Set Temp Alarm Threshold.

Sets the temperature alarm threshold to 40 through 100 degrees C inclusive. Byte 2 is a single unsigned binary byte; for example, 60 degrees C is encoded as 0x3C.

Command:

0	0x20	Unit ID
1	0x81	Message ID
2	0xnn	Temperature alarm threshold in degrees C, unsigned binary byte (40 to 100 inclusive)

Response:

ACK after a successful EEPROM write; NAK with error 100 (out of range) if the value is below 40 or above 100 degrees C, or error 101 (EE write failure) if the EEPROM write fails.

For a valid value, the threshold in RAM is updated before the EEPROM write. If saving fails, the new threshold remains active for the current session even though the command returns NAK error 101. A restart reloads the threshold from EEPROM.

4.17 Command 0x82: Set Multiplier Factor.

Sets the configured frequency multiplication factor on unit ID 0x20. The factor is saved to EEPROM and updated in RAM only after the EEPROM write succeeds.

Command:

0	0x20	Unit ID
1	0x82	Message ID
2	0x01	Multiplication factor: 0x01 = x1, 0x02 = x2, 0x04 = x4, 0x08 = x8

Response:

ACK after a successful EEPROM write and RAM update; NAK with error 100 (out of range) for an invalid factor, or error 101 (EE write failure) if saving fails. The RAM factor remains unchanged if the EEPROM write fails.

Rules:

Allowed multiplication factors are 1, 2, 4 and 8, encoded as a single unsigned binary byte: 0x01, 0x02, 0x04 or 0x08 respectively.

This command does not reconfigure the DDS or change the configured minimum and maximum output frequency limits. Set those limits separately if required, then send a new CW or sweep configuration to apply frequencies using the new multiplier factor.

4.18 Command 0x83: Set Min Frequency.

Sets the minimum output frequency limit on unit ID 0x20. The value is a 32-bit unsigned integer in kHz, transmitted least significant byte first. It is saved to EEPROM and updated in RAM only after the EEPROM write succeeds.

Command:

0	0x20	Unit ID
1	0x83	Message ID
2	0xC0	Minimum output frequency in kHz, 32-bit unsigned, least significant byte first. Example: 0x000927C0 = 600,000 kHz; bytes 0xC0 0x27 0x09 0x00.
3	0x27	
4	0x09	
5	0x00	

Response:

ACK after a successful EEPROM write and RAM update; NAK with error 100 (out of range) for a value below 1,000 or above 8,000,000 kHz, or error 101 (EE write failure) if saving fails. The RAM limit remains unchanged if the EEPROM write fails.

Rules:

The caller must ensure that the minimum frequency is below the configured maximum frequency. This command does not compare the two limits.

Minimum accepted value: 1,000 kHz (1 MHz), inclusive.

Maximum accepted value: 8,000,000 kHz (8 GHz), inclusive.

This command changes the stored and RAM frequency limit but does not retune the DDS. Send a new CW or sweep configuration to apply frequencies within the configured limits.

4.19 Command 0x84: Set Max Frequency.

Sets the maximum output frequency limit on unit ID 0x20. The value is a 32-bit unsigned integer in kHz, transmitted least significant byte first. It is saved to EEPROM and updated in RAM only after the EEPROM write succeeds.

Command:

0	0x20	Unit ID
1	0x84	Message ID
2	0x68	Maximum output frequency in kHz, 32-bit unsigned, least significant byte first. Example: 0x00098968 = 625,000 kHz; bytes 0x68 0x89 0x09 0x00.
3	0x89	
4	0x09	
5	0x00	

Response:

ACK after a successful EEPROM write and RAM update; NAK with error 100 (out of range) for a value below 1,000 or above 8,000,000 kHz, or error 101 (EE write failure) if saving fails. The RAM limit remains unchanged if the EEPROM write fails.

Rules:

The caller must ensure that the maximum frequency is above the configured minimum frequency. This command does not compare the two limits.

Minimum accepted value: 1,000 kHz (1 MHz), inclusive.

Maximum accepted value: 8,000,000 kHz (8 GHz), inclusive.

This command changes the stored and RAM frequency limit but does not retune the DDS. Send a new CW or sweep configuration to apply frequencies within the configured limits.

4.20 Command 0x85: Mute Output. (User)

Mutes or unmutes the DDS output on unit ID 0x20 in both CW and sweep modes. Control byte 0x31 selects Mute; 0x30 selects Unmute.

0	0x20	Unit ID
1	0x85	Message ID
2	0x31	0x30: Unmute / 0x31: Mute

Response:

ACK when the control byte is 0x31 (Mute) or 0x30 (Unmute); NAK with error 100 (out of range) for any other control byte.

Selecting Mute sets the mute warning bit; selecting Unmute clears it. The mute state is held in RAM and is not saved to EEPROM. It is initialised to unmuted after a restart.

4.21 Command 0x86: Set Output Level.

Sets the DDS output amplitude scale from 0 to 100 inclusive, encoded as a single unsigned binary byte. The value controls the DDS amplitude scale and is not an output power in dBm.

Command:

0	0x20	Unit ID
1	0x86	Message ID
2	0x4E	DDS amplitude scale, unsigned binary byte (0 to 100 inclusive). Example: 0x4E = 78 decimal.

Response:

ACK after the value is saved to EEPROM, updated in RAM and applied to the DDS. NAK with error 100 (out of range) if the value exceeds 100, or error 101 (EE write failure) if saving fails.

The setting is saved persistently and applied immediately after a successful EEPROM write; no restart is required. If the EEPROM write fails, the RAM level and DDS amplitude setting remain unchanged.

4.22 Command 0x90: Get DDS Pin Control State.

Reports the GPIO output-register states for the DDS External Power Down, OSK, DRHOLD and DRCTL control signals on unit ID 0x20. Each state is encoded as an ASCII byte: 0x30 = LOW and 0x31 = HIGH. These are the programmed output states, not measured pin levels; their control functions depend on the DDS configuration.

0	0x20	Unit ID
1	0x90	Message ID

Response:

0	0x20	Unit ID
1	0x90	Message ID
2	0x44	Data to follow
3	0x30	External Power Down, 0x30: LOW / 0x31: HIGH
4	0x31	OSK, 0x30: LOW / 0x31: HIGH
5	0x30	DRHOLD, 0x30: LOW / 0x31: HIGH
6	0x30	DRCTL, 0x30: LOW / 0x31: HIGH

Note: Some of these signals are changing dynamically.

4.23 Command 0x91: Pulse IO Update.

Generates a pulse on the DDS IO_UPDATE control line. The firmware drives the line HIGH, waits through a short delay loop, then drives it LOW.

No control byte is required. This command generates the IO_UPDATE pulse only; it does not transfer register data over the DDS data bus.

Command:

0	0x20	Unit ID
1	0x91	Message ID

Response:

ACK after the IO_UPDATE pulse is complete.

4.24 Command 0x92: DDS Master Reset.

Generates a pulse on the DDS MASTER_RESET control line. The firmware drives the line HIGH, waits through a short delay loop, then drives it LOW. No control byte is required.

The DDS must be reconfigured after this operation; this command does not reconfigure it automatically. The reset applies to the DDS, not the STM32.

Command:

0	0x20	Unit ID
1	0x92	Message ID

Response:

ACK after the MASTER_RESET pulse is complete.

4.25 Command 0x93: External Power Down.

Directly controls the active-high External Power Down pin on the DDS chip. Control byte 0x31 drives the pin HIGH, asserting power-down; 0x30 drives it LOW, releasing power-down. The pin state is not saved to EEPROM.

Command:

0	0x20	Unit ID
1	0x93	Message ID
2	0x31	0x31: HIGH (assert power-down) / 0x30: LOW (release power-down)

Response:

ACK after setting the pin when the control byte is 0x31 (HIGH) or 0x30 (LOW); NAK with error 100 (out of range) for any other control byte.

4.26 Command 0x94: External OSK.

Directly controls the external OSK pin on the DDS chip. Control byte 0x31 drives the pin HIGH; 0x30 drives it LOW. The pin state is not saved to EEPROM.

External Output Shift Keying, active high.

Command:

0	0x20	Unit ID
1	0x94	Message ID
2	0x31	0x31: HIGH / 0x30: LOW

Response:

ACK after setting the pin when the control byte is 0x31 (HIGH) or 0x30 (LOW); NAK with error 100 (out of range) for any other control byte.

This command changes OSK directly without updating the mute flag or mute warning bit. Mute/unmute commands and DDS configuration can subsequently change the OSK state.

4.27 Command 0x95: DR Hold.

Directly controls the DRHOLD pin on the DDS chip. Control byte 0x31 drives the pin HIGH, asserting ramp hold; 0x30 drives it LOW, releasing ramp hold. The pin state is not saved to EEPROM.

Digital Ramp Hold, active high, pauses the sweep when active.

Command:

0	0x20	Unit ID
1	0x95	Message ID
2	0x31	0x31: HIGH (assert ramp hold) / 0x30: LOW (release ramp hold)

Response:

ACK after setting the pin when the control byte is 0x31 (HIGH) or 0x30 (LOW); NAK with error 100 (out of range) for any other control byte.

Sweep-control code also changes DRHOLD. This command sets the pin directly but does not guarantee that the requested state remains held during subsequent sweep operation.

4.28 Command 0x96: DRCTL.

Directly controls the DRCTL pin on the DDS chip. Control byte 0x31 drives the pin HIGH; 0x30 drives it LOW. The pin state is not saved to EEPROM.

Digital Ramp Control: the effect of the HIGH or LOW state depends on the DDS ramp configuration.

Command:

0	0x20	Unit ID
1	0x96	Message ID
2	0x31	0x31: HIGH / 0x30: LOW

Response:

ACK after setting the pin when the control byte is 0x31 (HIGH) or 0x30 (LOW); NAK with error 100 (out of range) for any other control byte.

Sweep-control code also changes DRCTL. This command sets the pin directly but does not guarantee that the requested state remains held during subsequent sweep operation.

4.29 Command 0xA0: Get Status. (User)

Requests the stored status, alarm, warning and temperature values from unit ID 0x20. This command does not request a fresh temperature-sensor reading.

Command:

0	0x20	Unit ID
1	0xA0	Message ID

Response:

0	0x20	Unit ID
1	0xA0	Message ID
2	0x44	Data to follow
3	0x04	Overall Bitfield
4	0x00	Alarm Bitfield
5	0x00	Warning Bitfield
6	0x03	Status Bitfield
7	0x2A	Stored temperature, low byte of data.temp. Example: 0x2A = 42 degrees C; no fresh sensor read is performed by this command.

Overall Bitfield (0: Clear, 1: Set). Bit 2 is set when both status bits 0 and 1 are set, independently of alarms and warnings. It may be set together with overall alarm and warning bits. A value of 0x04 means both status bits are set and neither the overall alarm nor warning bit is set.

0	1: Alarm
1	1: Warning
2	1: Both status bits 0 and 1 are set (1V8 power good and PLL locked); independent of alarms and warnings
3-7	0 (Clear)

Alarm Bitfield: (0: Clear, 1: Set, 0x00==No Alarms)

0	EE I2C Comms
1	TMP100 I2C Comms
2	Over Temperature
3	PCB ID unknown on power up, config?
4	PLL not locked on start up
5	Sweep stalled
6	DROVER interrupt stalled
7	0 (Clear; unused)

Warning Bitfield: (0: Clear, 1: Set, 0x00==No Warnings)

0	Low Level Test Mode
1	Output Muted
2-7	0 (Clear)

Status Bitfield: (0: Clear, 1: Set, 0x03==Status is all good)

0	1V8 PG (Power Good)
1	PLL Locked
2-7	0 (Clear)

4.30 Command 0xA1: Clear Alarms. (User)

Clears all alarm bits in RAM, clears the warning bitfield, then rebuilds the test-mode and mute warning bits from data.lowLevelTestMode and data.mute respectively. Warnings for active conditions therefore remain reported. No control byte is required. Status bits, mute state and the low level test-mode flag are unchanged. Alarm conditions are not removed by this command; alarms may be raised again if their conditions persist.

Command:

0	0x20	Unit ID
1	0xA1	Message ID

Response:

ACK after clearing the alarm bitfield and rebuilding the warning bits from the current mute and test-mode states.

4.31 Command 0xA5: Set Unit ID.

This global command is sent to broadcast address 0xFF. Every connected slave accepts the new unit ID, which must be in the range 0x00 to 0xFE. The ID takes effect immediately without a restart. It is recommended to have only one unit connected when using this command, because all connected slaves will otherwise acquire the same ID.

The firmware updates the active ID in RAM, then attempts to save it to EEPROM. The EEPROM write result is not checked. If saving fails, the new ID remains active for the current session but is not guaranteed to persist after a restart.

Command:

0	0xFF	Unit ID (Global)
1	0xA5	Message ID
2	0x20	New unit ID, unsigned binary byte: 0x00 to 0xFE (0xFF is reserved for broadcast)

Response: None. Slaves must not send ACK or NAK in response to this global command, because simultaneous replies would clash on the bus. EEPROM write failures also remain silent.

4.32 Command 0xB0: Get Build Info. (User)

Requests build information from unit ID 0x20. Index 0 selects the main DDS board; index 1 selects the multiplier board. Other indices return NAK with error 107 (index out of range).

0	0x20	Unit ID
1	0xB0	Message ID
2	0x00	Index: 0 = Main DDS board / 1 = Multiplier board

Response:

0	0x20	Unit ID

1	0xB0	Message ID
2	0x44	Data to follow
3	0x00	Index: 0 = Main DDS board / 1 = Multiplier board
4	0x45	PCB ID string: EJWM0013AA
5	0x4A	
6	0x57	
7	0x4D	
8	0x30	
9	0x30	
10	0x31	
11	0x33	
12	0x41	
13	0x41	
14	0x1F	PCB Version: 0x011F = 287
15	0x01	Max: 999
16	0x45	PCA ID string: EGRD0024AA
17	0x47	
18	0x52	
19	0x44	
20	0x30	
21	0x30	
22	0x32	
23	0x34	
24	0x41	
25	0x41	
26	0x23	PCA Version: 0x0123 = 291
27	0x01	Max: 999
28	0x45	SN string: ESK0000026
29	0x53	
30	0x4B	
31	0x30	
32	0x30	
33	0x30	
34	0x30	
35	0x30	
36	0x32	
37	0x36	
38	0x45	Application firmware ID: EVFP0003AA
39	0x56	(10 bytes; main DDS board only)
40	0x46	
41	0x50	
42	0x30	
43	0x30	
44	0x30	
45	0x33	
46	0x41	
47	0x41	
48	0x14	Application firmware version, 16-bit
49	0x00	unsigned, LSB first. Current release

		example: 0x0014 = 20 (main DDS board only).
50	0x5A	Application protocol ID: ZPAG0002AA (10 bytes; main DDS board only)
51	0x50	
52	0x41	
53	0x47	
54	0x30	
55	0x30	
56	0x30	
57	0x32	
58	0x41	
59	0x41	
60	0x0C	Application protocol version, 16-bit unsigned, LSB first. Current example: 0x000C = 12 (main DDS board only).
61	0x00	
62	0xnn	Bootloader firmware ID (10 bytes; main DDS board only). Value depends on the bootloader information available to the application.
63	0xnn	
64	0xnn	
65	0xnn	
66	0xnn	
67	0xnn	
68	0xnn	
69	0xnn	
70	0xnn	
71	0xnn	
72	0xnn	Bootloader firmware version, 16-bit unsigned, LSB first (main DDS board only). Value depends on available bootloader information.
73	0xnn	
74	0xnn	Bootloader protocol ID (10 bytes; main DDS board only). Value depends on available bootloader information.
75	0xnn	
76	0xnn	
77	0xnn	
78	0xnn	
79	0xnn	
80	0xnn	
81	0xnn	
82	0xnn	
83	0xnn	
84	0xnn	Bootloader protocol version, 16-bit unsigned, LSB first (main DDS board only). Value depends on available bootloader information.
85	0xnn	

The response table shows the main DDS board layout (index 0). Bytes 4 to 37 contain PCB ID/version, PCA ID/version and serial number for either board. For index 1, append the CRC immediately after byte 37; no application or bootloader firmware/protocol fields follow.

All ID and serial-number strings occupy exactly 10 bytes, without an appended NUL terminator. All version fields occupy two bytes, least significant byte first. The PCB and PCA

versions are limited to 999 by their set commands. Application and bootloader version fields are returned as stored 16-bit values.

For index 0, append the CRC after byte 85: 86 message bytes plus two CRC bytes. For index 1, the response is 38 message bytes plus two CRC bytes. Apply byte substitution and frame markers as described in section 2.

PCB, PCA and serial-number values shown are illustrative. The application currently identifies itself as EVFP0003AA release 20 and protocol ZPAG0002AA version 12. These values may change with firmware releases. The 0xnn entries represent variable bootloader bytes, not literal transmitted values.

4.33 Command 0xB1: Set PCB ID.

Sets the PCB ID for the selected board: index 0 is the main DDS board and index 1 is the multiplier board. Bytes 3 to 12 contain exactly 10 ASCII characters, without an appended NUL terminator. The required format is four uppercase letters A-Z, four digits 0-9, then two uppercase letters A-Z (AAAA9999AA). The example EORF0013AA satisfies this format. The firmware validates the format only; it does not check the ID against a list of recognised product codes.

Command:

0	0x20	Unit ID
1	0xB1	Message ID
2	0x00	Index: 0 = Main DDS board / 1 = Multiplier board
3	0x45	‘E’
4	0x4F	‘O’
5	0x52	‘R’
6	0x46	‘F’
7	0x30	‘0’
8	0x30	‘0’
9	0x31	‘1’
10	0x33	‘3’
11	0x41	‘A’
12	0x41	‘A’

Response:

ACK after the PCB ID is saved to EEPROM. NAK with error 107 (index out of range) for an unsupported board index, error 100 (out of range) for an invalid ID format, or error 101 (EE write failure) if saving fails.

For a valid ID, the selected board's PCB ID in RAM is updated before the EEPROM write. If saving fails, the new ID remains active in RAM even though the command returns NAK error 101. It is not guaranteed to persist after a restart.

4.34 Command 0xB2: Set PCB Version.

Sets the PCB version for the selected board: index 0 is the main DDS board and index 1 is the multiplier board. The version is a 16-bit unsigned binary value, transmitted least significant byte first, with an accepted range of 0 to 999 inclusive.

Command:

0	0x20	Unit ID
1	0xB2	Message ID
2	0x00	Index: 0 = Main DDS board / 1 = Multiplier board
3	0x03	PCB version, 16-bit unsigned, least significant byte first. Example: 0x0003 = 3, transmitted as 0x03 0x00. Valid range: 0 to 999 inclusive.
4	0x00	

Response:

ACK after the PCB version is saved to EEPROM. NAK with error 100 (out of range) for a version above 999, error 101 (EE write failure) if saving fails, or error 107 (index out of range) for an unsupported board index.

For a valid version, the selected board's PCB version in RAM is updated before the EEPROM write. If saving fails, the new version remains active in RAM even though the command returns NAK error 101. It is not guaranteed to persist after a restart.

4.35 Command 0xB3: Set Serial Number String.

Sets the serial number for the selected board: index 0 is the main DDS board and index 1 is the multiplier board. Bytes 3 to 12 contain exactly 10 ASCII characters, without an appended NUL terminator. The required format is three uppercase letters A-Z followed by seven digits 0-9 (AAA9999999). The example ESK0000026 satisfies this format.

Command:

0	0x20	Unit ID
1	0xB3	Message ID

2	0x00	Index: 0 = Main DDS board / 1 = Multiplier board
3	0x45	Serial Number ESK0000026
4	0x53	
5	0x4B	
6	0x30	
7	0x30	
8	0x30	
9	0x30	
10	0x30	
11	0x32	
12	0x36	

Response:

ACK after the serial number is saved to EEPROM. NAK with error 100 (out of range) for an invalid serial-number format, error 101 (EE write failure) if saving fails, or error 107 (index out of range) for an unsupported board index.

For a valid serial number, the selected board's serial number in RAM is updated before the EEPROM write. If saving fails, the new serial number remains active in RAM even though the command returns NAK error 101. It is not guaranteed to persist after a restart.

4.36 Command 0xB4: Set PCA ID.

Sets the PCA ID for the selected board: index 0 is the main DDS board and index 1 is the multiplier board. Bytes 3 to 12 contain exactly 10 ASCII characters, without an appended NUL terminator. The required format is four uppercase letters A-Z, four digits 0-9, then two uppercase letters A-Z (AAAA9999AA). The example EGRD0024AA satisfies this format. The firmware validates the format only; it does not check the ID against a list of recognised product codes.

Command:

0	0x20	Unit ID
1	0xB4	Message ID
2	0x00	Index: 0 = Main DDS board / 1 = Multiplier board
3	0x45	‘E’
4	0x47	‘G’
5	0x52	‘R’
6	0x44	‘D’
7	0x30	‘0’
8	0x30	‘0’
9	0x32	‘2’
10	0x34	‘4’
11	0x41	‘A’
12	0x41	‘A’

Response:

ACK after the PCA ID is saved to EEPROM. NAK with error 107 (index out of range) for an unsupported board index, error 100 (out of range) for an invalid ID format, or error 101 (EE write failure) if saving fails.

For a valid ID, the selected board's PCA ID in RAM is updated before the EEPROM write. If saving fails, the new ID remains active in RAM even though the command returns NAK error 101. It is not guaranteed to persist after a restart.

4.37 Command 0xB5: Set PCA Version.

Sets the PCA version for the selected board: index 0 is the main DDS board and index 1 is the multiplier board. The version is a 16-bit unsigned binary value, transmitted least significant byte first, with an accepted range of 0 to 999 inclusive.

Command:

0	0x20	Unit ID
1	0xB5	Message ID
2	0x00	Index: 0 = Main DDS board / 1 = Multiplier board
3	0x05	PCA version, 16-bit unsigned, least significant byte first. Example: 0x0005 = 5, transmitted as 0x05 0x00. Valid range: 0 to 999 inclusive.
4	0x00	

Response:

ACK after the PCA version is saved to EEPROM. NAK with error 100 (out of range) for a version above 999, error 101 (EE write failure) if saving fails, or error 107 (index out of range) for an unsupported board index.

For a valid version, the selected board's PCA version in RAM is updated before the EEPROM write. If saving fails, the new version remains active in RAM even though the command returns NAK error 101. It is not guaranteed to persist after a restart.

4.38 Command 0xC0: Get Config. (User)

Returns the current RAM configuration and build information from unit ID 0x20. This command does not reread EEPROM. The example uses an x4 frequency multiplier with output limits of 2,000,000 to 4,000,000 kHz (2 to 4 GHz), consistent with the sweep example in section 4.15.2.

Command:

0	0x20	Unit ID
1	0xC0	Message ID

Response:

Byte 130 identifies the current operating mode: 0 = CW, 1 = Sweep. Byte 192 selects modulation mode and byte 193 contains the packed modulation settings.

0	0x20	Unit ID
1	0xC0	Message ID
2	0x44	Data to follow
3	0x45	PCB ID: EORF0013AA
4	0x4F	
5	0x52	
6	0x46	
7	0x30	
8	0x30	
9	0x31	
10	0x33	
11	0x41	
12	0x41	
13	0x03	PCB Version: 0x0003 = 3
14	0x00	Max: 999
15	0x45	PCA ID: EGRD0024AA
16	0x47	
17	0x52	
18	0x44	
19	0x30	
20	0x30	
21	0x32	
22	0x34	
23	0x41	
24	0x41	
25	0x05	PCA Version: 0x0005 = 5
26	0x00	Max: 999
27	0x45	SN string: ESK0000026
28	0x53	
29	0x4B	
30	0x30	
31	0x30	
32	0x30	
33	0x30	
34	0x30	
35	0x32	
36	0x36	

37	0x45	Application firmware ID: EVFP0003AA (10 bytes)
38	0x56	
39	0x46	
40	0x50	
41	0x30	
42	0x30	
43	0x30	
44	0x33	
45	0x41	
46	0x41	
47	0x14	Application firmware version, 16-bit unsigned, LSB first. Current example: 0x0014 = 20.
48	0x00	
49	0x5A	Application protocol ID: ZPAG0002AA (10 bytes)
50	0x50	
51	0x41	
52	0x47	
53	0x30	
54	0x30	
55	0x30	
56	0x32	
57	0x41	
58	0x41	
59	0x0C	Application protocol version, 16-bit unsigned, LSB first. Current example: 0x000C = 12.
60	0x00	
61	0xnn	Bootloader firmware ID (10 bytes); variable bootloader information.
62	0xnn	
63	0xnn	
64	0xnn	
65	0xnn	
66	0xnn	
67	0xnn	
68	0xnn	
69	0xnn	
70	0xnn	
71	0xnn	Bootloader firmware version, 16-bit unsigned, LSB first; variable bootloader information.
72	0xnn	
73	0xnn	Bootloader protocol ID (10 bytes); variable bootloader information.
74	0xnn	
75	0xnn	
76	0xnn	
77	0xnn	
78	0xnn	
79	0xnn	
80	0xnn	
81	0xnn	
82	0xnn	
83	0xnn	

84	0xnn	Bootloader protocol version, 16-bit unsigned, LSB first; variable bootloader information.
85	0xnn	Multiplier-board PCB ID (10 bytes); value depends on the configured board.
86	0xnn	
87	0xnn	
88	0xnn	
89	0xnn	
90	0xnn	
91	0xnn	
92	0xnn	
93	0xnn	
94	0xnn	
95	0xnn	Multiplier-board PCB version, 16-bit unsigned, LSB first.
96	0xnn	
97	0xnn	Multiplier-board PCA ID (10 bytes); value depends on the configured board.
98	0xnn	
99	0xnn	
100	0xnn	
101	0xnn	
102	0xnn	
103	0xnn	
104	0xnn	
105	0xnn	
106	0xnn	
107	0xnn	Multiplier-board PCA version, 16-bit unsigned, LSB first.
108	0xnn	
109	0xnn	Multiplier-board serial number (10 bytes).
110	0xnn	
111	0xnn	
112	0xnn	
113	0xnn	
114	0xnn	
115	0xnn	
116	0xnn	
117	0xnn	
118	0xnn	
119	0x04	Frequency multiplier: example 0x04 = x4. Supported factors: 1, 2, 4 and 8.
120	0x80	Minimum output frequency, 32-bit unsigned kHz, LSB first. Example: 0x001E8480 = 2,000,000 kHz.
121	0x84	
122	0x1E	
123	0x00	
124	0x00	Maximum output frequency, 32-bit unsigned kHz, LSB first. Example: 0x003D0900 = 4,000,000 kHz.
125	0x09	
126	0x3D	
127	0x00	
128	0x3C	Temp Alarm Threshold, 0x3C = 60 deg. C

129	0x4E	DDS amplitude scale, unsigned binary byte (0 to 100). Example: 0x4E = 78; not dBm.
130	0x01	Operating mode: 0 = CW, 1 = Sweep.
131	0x50	CW Frequency 0x256250 2450,000 KHz
132	0x62	
133	0x25	
134	0x00	
135	0x03	Number of active sweeps: 0 to 6; example 3. All six frequency pairs are returned.
136	0x00	Sweep 1 Start Frequency 0x249F00 2400,000 KHz
137	0x9F	
138	0x24	
139	0x00	
140	0x20	Sweep 1 Stop Frequency 0x24ED20 2420,000 KHz
141	0xED	
142	0x24	
143	0x00	
144	0x40	Sweep 2 Start Frequency 0x253B40 2440,000 KHz
145	0x3B	
146	0x25	
147	0x00	
148	0x60	Sweep 2 Stop Frequency 0x258960 2460,000 KHz
149	0x89	
150	0x25	
151	0x00	
152	0x80	Sweep 3 Start Frequency 0x25D780 2480,000 KHz
153	0xD7	
154	0x25	
155	0x00	
156	0xA0	Sweep 3 Stop Frequency 0x2625A0 2500,000 KHz
157	0x25	
158	0x26	
159	0x00	
160	0x00	Sweep 4 Start Frequency
161	0x00	
162	0x00	
163	0x00	
164	0x00	Sweep 4 Stop Frequency
165	0x00	
166	0x00	
167	0x00	
168	0x00	Sweep 5 Start Frequency
169	0x00	
170	0x00	
171	0x00	
172	0x00	Sweep 5 Stop Frequency
173	0x00	
174	0x00	
175	0x00	

176	0x00	Sweep 6 Start Frequency
177	0x00	
178	0x00	
179	0x00	
180	0x00	Sweep 6 Stop Frequency
181	0x00	
182	0x00	
183	0x00	
184	0x88	Sweep frequency step, 32-bit unsigned Hz, LSB first. Example: 0x00001388 = 5,000 Hz.
185	0x13	
186	0x00	
187	0x00	
188	0x14	Ramp Rate, 32-bit unsigned, LSB first. Example: 0x00000014 = 20 DDS ramp-rate counts; not nanoseconds.
189	0x00	
190	0x00	
191	0x00	
192	0x00	Modulation mode: 0 = None, 1 = Dither, 2 = Injected.
193	0x20	Packed modulation settings: bits 1-0 = time; bits 3-2 = pattern; bits 5-4 = modulation frequency. Example 0x20 = 1 s, pattern OFF, 100 kHz.

All ID and serial-number strings are fixed 10-byte fields without an appended NUL terminator. Versions are 16-bit unsigned values; frequency, step and Ramp Rate fields are 32-bit unsigned values. All multibyte numeric fields are transmitted least significant byte first.

CW and sweep start/stop frequencies are output frequencies in kHz. The firmware multiplies the stored DDS frequencies by the configured frequency multiplier when building this response. The sweep step is in Hz, and Ramp Rate is a count. Both CW and sweep fields are included regardless of the operating mode.

Byte 193 contains the following packed fields, using the format described in section 4.15.1. Bits 7-6 are zero.

Bits 1-0	**Modulation time**
0	1 s
1	100 ms
2	100 us
3	Random

Bits 3-2	**Pattern**
0	OFF
1	PRBS3
2	PRBS5
3	PRBS7

Bits 5-4	Modulation frequency
0	OFF
1	75 kHz
2	100 kHz
3	150 kHz

Each value applies to its two-bit field, rather than the whole byte.

The response has 194 message bytes (offsets 0 to 193). Append the CRC at offsets 194 and 195, giving 196 decoded bytes including CRC. Byte substitution and frame markers are applied as described in section 2.

The table values are illustrative configuration/build data, not guaranteed defaults. The current application firmware example is EVFP0003AA release 20, with protocol ZPAG0002AA version 12. The 0xnn entries represent variable bytes, not literal transmitted values. A response-buffer overflow returns NAK error 100.

4.39 Command 0xE0: STM32 Soft Reset. (User)

Requests an STM32 system reset on unit ID 0x20 by calling NVIC_SystemReset(). No control byte is required. The unit resets immediately. During startup, configuration and the unit ID are reloaded from EEPROM; unsaved RAM changes are lost.

Command:

0	0x20	Unit ID
1	0xE0	Message ID

Response: None. The unit resets without sending ACK or NAK.

4.40 Command 0xF0: Boot Load Enable. (User)

Requests bootloader entry for unit ID 0x20. The firmware writes 0xA5 to the EEPROM bootloader-enable flag at address 0x0000 (EE_ADDR_BL_EN), then requests an STM32 system reset using NVIC_SystemReset(). No control byte or additional payload is required.

Command:

0	0x20	Unit ID
1	0xF0	Message ID

Response:

On a successful EEPROM write, the STM32 resets without sending ACK or DATA. If the write fails, the unit returns NAK with error 101 (EE write failure) and does not perform the reset.

The command requests bootloader entry through the stored enable flag. The subsequent bootloader behaviour and firmware-download protocol are outside the scope of this application-protocol command.

5 Document Change History

Release	Date	Notes
0.1	16/04/2024	Initial Release
0.2	26/04/2024	Added command 0xF0, Get Multiplier Info. Added command 0x50, Set default test settings. Added command 0x51, Erase EE, write 0xFF to all addresses. Added response type byte ACK, NAK or DATA to responses. Added Command ID to ACK and NAK responses. Added command 0x64, Set Low Level Test Mode. Added command 0x72, Test DDS Registers. Added barcodes EJWM0000AA to EJWM0014AA Improved example in section 2.4
0.3	29/04/2024	Added warning bit field to 0xA0 response. Added mute output message 0xD1. Improved test/dev barcode description of use.
0.4	29/04/2024	Increased size of version, 0->999 Added Set multiplier Info command 0x81. Changed Get multiplier Info command 0xF0 -> 0x80. Changed Mute command ID to 0x82. Removed redundant barcode section.
0.5	08/05/2024	Added command 0x97, Set Output Level. Added command 0x98, Get Output Level.

		Added index byte to command 0xB0, Get Build Info. Added index byte to command 0xB1, Set Bar Code String. Added index byte to command 0xB2, Set Version. Added index byte to command 0xB3, Set Serial Number String. Added command 0xC2, Get Sweep Info.
0.6	13/05/2024	Command 0xC0, Get Full Config, bytes added to response. Changed Mute command ID to 0x85. Added command 0x82, Set Mult Factor. Added command 0x83, Set Min Freq. Added command 0x84, Set Max Freq. Removed commands 0x80, 0x81, 0x98, 0xC2, 0xC5. Command 0xC1 is now 0x80, Set Sweep Config. Command 0xC6 is now 0x81, Set Temp Alarm Threshold. Command 0x97 is now 0x86, Set Output Level.
0.7	22/05/2024	Changed response to SetDefaultEE Command 0x50. Changed response to EraseEE Command 0x51.
0.8	7/10/2024	Added multi-sweep to SetSweepConfig Command 0x80 and GetConfig Command 0xC0.
0.9	20/10/2024	Corrected byte index in GetConfig Command 0xC0
10	9/11/2024	Added no sweep option, sweep count = 0 Modified error codes Removed single sweep Removed double sweep Multi sweep now referred to as sweep Mode modified, 0: CW, 1: Sweep
11	13/05/2026	Replaced Barcode with PCB ID string Replaced Version with PCB Version Added PCA ID String and Version. Relaced FW VAR+VER+BLD bytes with FW ID String and Version. This resulted in modifications to Get Build Info 0xB0 response. This resulted in modifications to Get Config 0xC0 response. New Command Set PCA ID string 0xB4 added. New Command Set PCA Version 0xB5 added. Renamed ZPAG0002AA-V11
12	28/10/2026	Updated the protocol to match FW V20, including CRC/framing, commands, defaults, modulation, build/configuration responses and error/status tables.

End of Document.